Predicting Conflict in Africa:

A Machine Learning Approach Using Environmental Stressors

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Abstract

This paper investigates the predictability of violent conflict in Africa using machine learning on satellite-derived environmental and socioeconomic data for 50km × 50km grid cells (2019–2023). We compare logistic regression, k-nearest neighbors, random forest, and neural networks, trained on climate, wealth, and historical conflict features to predict binary conflict occurrence. A conservative weighted ensemble achieves approximately 72% recall and 52% precision on the minority (conflict) class, illustrating both the value and the precision–recall tradeoffs of early-warning style models for policy and humanitarian audiences.

1 Introduction

Conflict remains a major barrier to development and stability, particularly where environmental stress and resource scarcity interact with inequality. Predicting where violence is more likely can help allocate aid and diplomatic attention—but models must balance catching true conflicts (recall) against false alarms (precision). Africa offers a demanding testbed: heterogeneous geography, uneven data quality, and strong class imbalance (most grid-years are non-conflict).

Research question. *Can the occurrence of violence in a given geography be predicted using local socioeconomic and environmental data?*

Key references (related forecasting & ML for conflict)

Hegre et al. 2024 describe the **VIEWS** system (Uppsala/PRIO), which produces grid-level fatality forecasts using gradient-boosted trees and related methods. Perry 2013 discusses whether machine learning can support conflict prediction in policy contexts. For k-NN background, see the course notes in 4780 (2017).

2 Data and outcomes

2.1 Sources

We combine: **UCDP** (georeferenced events); **ERA5** (climate); **Meta RWI** (wealth); **Hansen** forest change; **VIIRS** nighttime lights. Grid cells are 50km; labels indicate at least one conflict event in a grid-year.

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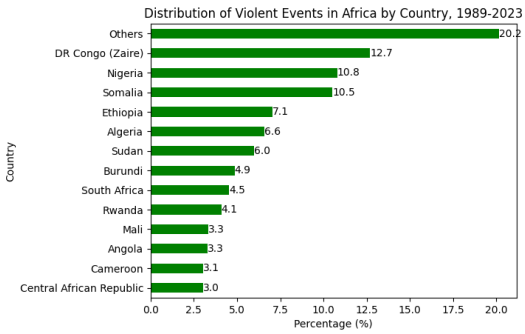
2.2 Descriptive patterns

Figure 1a shows the time series of violent events in Africa (1989–2023); Figure 1b shows concentration across countries.

(a) Conflicts over time



(b) Top countries

Figure 1: Descriptive conflict patterns (Africa)

2.3 Features and sample

The training file has 34,015 grid-year rows (2019–2023). Features include climate (temperature, precipitation, pressure), forest loss, lights, RWI, lags, and accumulated past conflict. The outcome is severely imbalanced ($\sim 92\%$ non-conflict); we use SMOTE during training for several models.

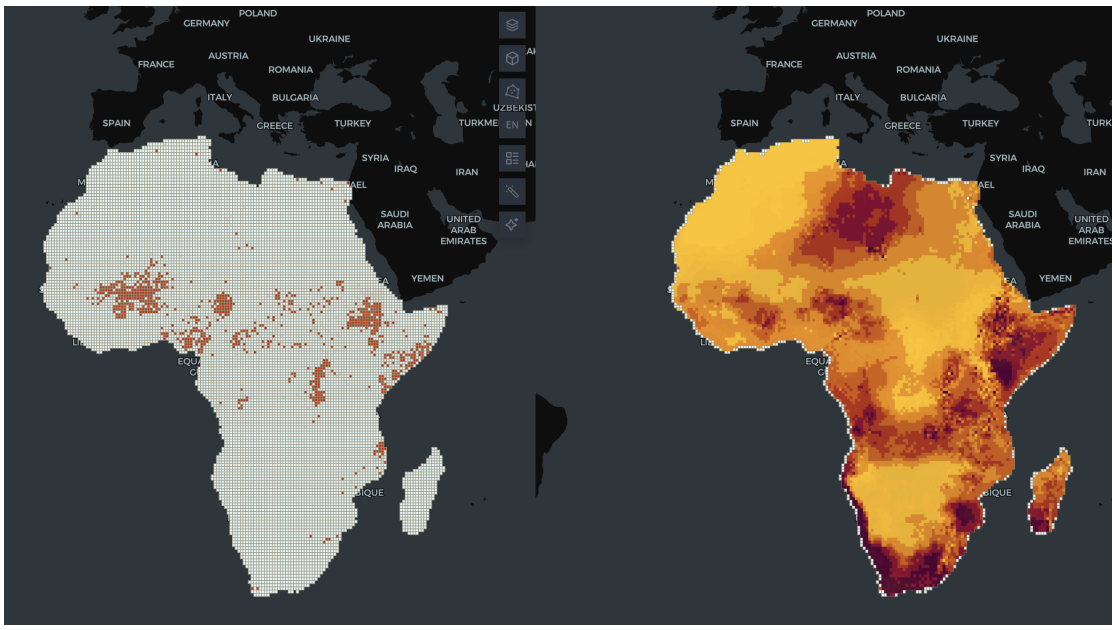


Figure 2: Spatial view of conflict aggregation and temperature anomalies (example year).

3 Models

3.1 Logistic regression (Elastic Net)

We estimate $P(y_i{=}1 \mid x_i) = \sigma(w^\top x_i + b)$ with elastic-net penalty on w. Features are standardized; hyperparameters (α, C) are tuned via cross-validation. SMOTE balances the training set.

3.2 k-nearest neighbors

We classify each grid using distance-weighted majority vote over k neighbors (cosine distance, $k{=}3$ in the selected specification).

3.3 Random forest

An ensemble of bagged trees with random feature subsets at each split; we report out-of-sample metrics and permutation-based feature importance.

3.4 Neural network

A small feedforward net ($64 \to 32 \to 1$, ReLU/sigmoid) trained with BCE and SGD.

3.5 Ensemble

We combine model probabilities with nonnegative weights summing to one and threshold 0.5. The **Conservative** weighting (LR 30%, KNN 25%, RF 35%, NN 10%) balances accuracy and recall for deployment-oriented use cases.

4 Results

4.1 Confusion matrices

Figures 3a–4b show confusion matrices for each classifier. File names `confusion_LR.png` and `confusion_KNN.png` should be exported from the training notebooks into `output/`; RF and NN use the repository filenames `confusion_RF.png` and `confusion_NN.png` (symlinked to the saved figures where spellings differ).

Missing figure

`../output/confusion_LR.png`

Run the logistic regression training notebook and save this name under `output/`.

(a) Logistic regression

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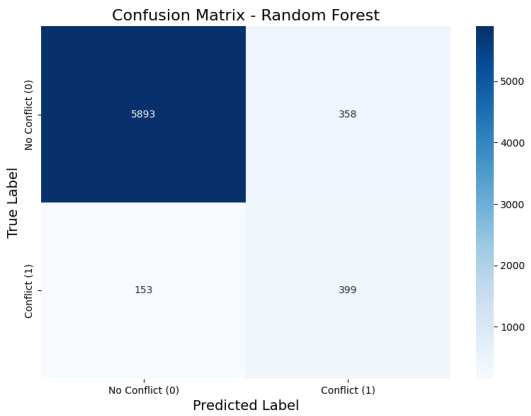
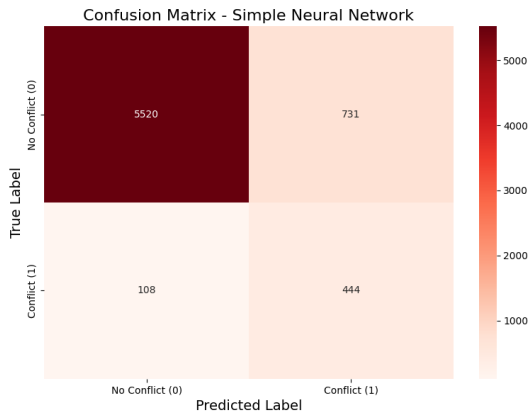
`../output/confusion_KNN.png`

Run the KNN notebook and save under `output/`.

(b) k-NN



(a) Random forest



(b) Neural network

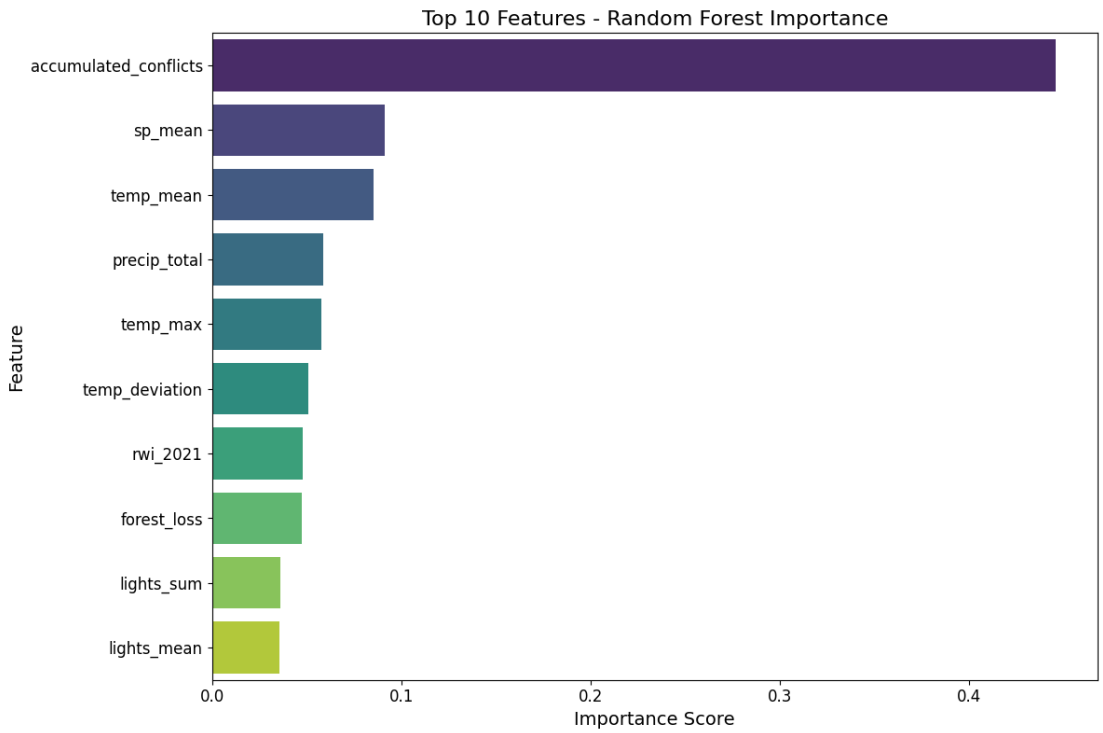
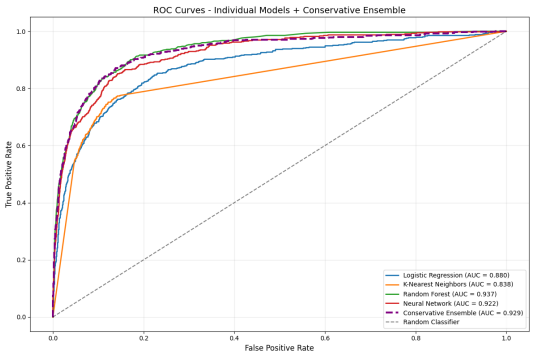
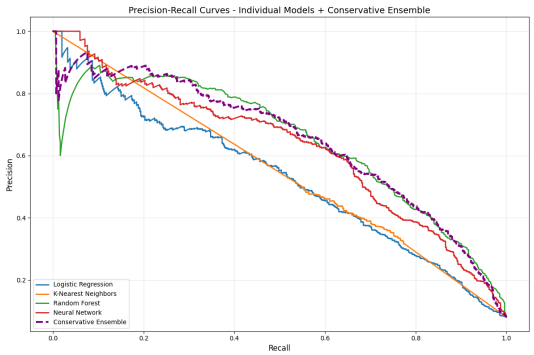


Figure 5: Random forest variable importance (impurity-based).

4.2 ROC and precision–recall



(a) ROC curves



(b) Precision–recall

Table 1: Illustrative ensemble comparison (test set; see notebook for full grid).

Scheme	Accuracy	Precision	Recall	F1
Conservative	0.923	0.520	0.723	0.605
RF-dominant	0.927	0.540	0.705	0.611

5 Discussion

Grid-level conflict prediction combines genuine signal with hard tradeoffs: high recall often implies many false positives, which can erode trust among practitioners. Ensembles help but do not remove the need for domain judgment and ethical safeguards if scores inform allocation or enforcement.

6 Conclusion

Environmental and socioeconomic features contain information about conflict risk at subnational scale, but models should be validated out-of-sample, communicated with uncertainty, and paired with governance discussion—especially where scores could be misused. Future work includes richer institutional covariates, better missing-data strategies for remote sensing, and tighter links to operational early-warning workflows.

Replication. Code and data pipeline: `https://github.com/anfelipecb/conflict-prediction-ml`. The course site hosts an interactive Kepler viewer under `docs/`.

References

4780, Cornell CS (2017). *Lecture notes: k-NN.* `https://www.cs.cornell.edu/courses/cs4780/2017sp/lectures/lecturenote02_kNN.html`.

Hegre, Håvard, Sofia Nordenving, and James Dale (2024). *Models in VIEWS, version Fatalities002.* Tech. rep. Violence and Impacts Early Warning System. Uppsala University and PRIO.

Perry, Chris (2013). *Can machines learn to predict a violent conflict?*